7. For the mathematically inclined reader, the statement here is that a beam of light, or massless particles more generally, can travel from any point in the interior of antideSitter space to spatial infinity and back, in finite time.

8. For the mathematically inclined reader, Maldacena worked in the context of $AdS_5 \times S^5$, with the boundary theory arising from the boundary of AdS_5.

9. This statement is more one of sociology than of physics. String theory grew out of the tradition of quantum particle physics, while loop quantum gravity grew out of the tradition of general relativity. However, it is important to note that, as of today, only string theory can make contact with the successful predictions of general relativity, since only string theory convincingly reduces to general relativity on large distance scales. Loop quantum gravity is understood well in the quantum domain, but bridging the gap to large-scale phenomena has proven difficult.

10. More precisely, as discussed further in Chapter 13 of *The Elegant Universe,* we have known how much entropy black holes contain since the work of Bekenstein and Hawking in the 1970s. However, the approach those researchers used was rather indirect, and never identified microscopic rearrangements—as in Chapter 6—that would account for the entropy they found. In the mid-1990s, this gap was filled by two string theorists, Andrew Strominger and Cumrun Vafa, who cleverly found a relation between black holes and certain configurations of branes in string/M-theory. Roughly, they were able to establish that certain special black holes would admit exactly the same number of rearrangements of their basic ingredients (whatever those ingredients might be) as do particular, special combinations of branes. When they counted the number of such brane rearrangements (and took the logarithm) the answer they found was the area of the corresponding black hole, in Planck units, divided by 4—exactly the answer for black hole entropy that had been found years before. In loop quantum gravity, researchers have also been able to show that the entropy of a black hole is proportional to its surface area, but getting the exact answer (surface area in Planck units divided by 4) has proven more of a challenge. If a particular parameter, known as the Immirzi parameter, is chosen